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Updated: October 25, 2026
Course: **Mathematics**

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[1]The author thanks Maksim Smirnov for providing materials for the course.

1 Basic topology

1.1 Metric spaces

Definition 1.1. A set X, whose elements we shall call *points*, is said to be *a metric space* if with any two points p and q of X there is associated a real number $d(p,q)$, called the *distance* from p to q, such that

(a) $d(p,q) > 0$ if $p \neq q$ and $d(p,p) = 0$,

(b) $d(p,q) = d(q,p)$,

(c) $d(p,q) \leq d(p,r) + d(r,q)$, for $\forall r \in X$.

Any function with these three properties is called a *distance function*, or *a metric*.

Example 1.2 (Metric spaces)**.** The following are examples of the metric spaces:

1. the set of real numbers $\mathbb{R}$ with a metric $d(p,q) = |p-q|$,
2. a real plane $\mathbb{R}^2$ with a metric $d(\mathbf{p},\mathbf{q}) = \sqrt{(p_1-q_1)^2 + (p_2-q_2)^2} := \|\mathbf{p}-\mathbf{q}\|$ (Eucledian distance),
3. a real plane $\mathbb{R}^2$ with a metric $d(\mathbf{p},\mathbf{q}) = |p_1-q_1| + |p_2-q_2|$ (Manhattan distance),
4. the set of probability distributions defined on the same measurable space with a metric $d(P,Q) = \frac{1}{\sqrt{2}}\left(\int\left(\sqrt{p(x)} - \sqrt{q(x)}\right)^2 dx\right)^{1/2}$ (Hellinger distance).

It is important to observe that every subset Y of a metric space X is a metric space in its own right, with the same distance function. Thus, every subset of a Euclidean space is a metric space.

Definition 1.3. By the *segment* (a,b) we mean the set of all real numbers x such that $a < x < b$. By the *interval* $[a,b]$ we mean the set of all real numbers x such that $a \leq x \leq b$.

If $a_i < b_i$ for $i = 1,\ldots,k$, the set of all points $\mathbf{x} = (x_1,\ldots,x_k)$ in $\mathbb{R}^k$ whose coordinates satisfy the inequalities $a_i \leq x_i \leq b_i$ $(1 \leq i \leq k)$ is called a *k-cell*. Thus, a 1-cell is an interval, a 2-cell is a rectangle, etc.

If $\mathbf{x} \in \mathbb{R}^k$ and $r > 0$, the *open* (or *closed*) *ball* B with center at $\mathbf{x}$ and radius r is defined to be the set of all $\mathbf{y} \in \mathbb{R}^k$ such that $\|\mathbf{y}-\mathbf{x}\| < r$ (or $\|\mathbf{y}-\mathbf{x}\| \leq r$).

We call a set $E \subset \mathbb{R}^k$ *convex* if

$$\lambda\mathbf{x} + (1-\lambda)\mathbf{y} \in E$$

whenever $\mathbf{x} \in E$, $\mathbf{y} \in E$, and $0 < \lambda < 1$. For example, balls are convex. It is also easy to see that k-cells are convex.

Definition 1.4. Let X be a metric space. All points and sets mentioned below are understood to be elements and subsets of X.

(a) *A neighborhood* of a point p is a set $N_r(p)$ consisting of all points q such that $d(p,q) < r$. The number r is called the *radius* of $N_r(p)$.

(b) A point p is a *limit point* of the set E if *every* neighborhood of p contains a point $q \neq p$ such that $q \in E$. Example: take a set $A := (0,1)$. Point 0 is a limit point, because any open interval, say $(-\varepsilon,\varepsilon)$, intersects A.

(c) If $p \in E$ and p is not a limit point of E, then p is called an *isolated point* of E. Example: take a set $A = \{n^{-1} : n \in \mathbb{N}\}$. Each element is an isolated point because you can take a small interval around n^{-1} that avoids the other fractions in the set.

(d) E is *closed* if every limit point of E is a point of E. Example: take $A = [0,1]$. Both 0 and 1 are limit points and both belong to the set A. A set $B = (0,1]$ is not closed because a limit point 0 does not belong to the set.

(e) A point p is an *interior point* of E if there is a neighborhood $N_r(p)$ of p such that $N \subset E$. Example: take a set $A = (0,1)$. A point 0.5 is an interior point because there is a neighborhood around it, say, $N_{0.1}(0.5)$ that belongs to the set A; if $N_{0.1}(0.5) = (0.4,0.6) := B$, we have $B \subset A$. On the other hand, if $C = [0.5,1]$, 0.5 is not an interior point of C, because there is no neighborhood around it that is a subset of C; some points of that neighborhood are outside of C.

(f) E is *open* if every point of E is an interior point of E.

(g) The *complement* of E (denoted by E^c) is the set of all points $p \in X$ such that $p \notin E$.

(h) E is *perfect* if E is closed and if every point of E is a limit point of E. Example: take $A = [0, 1]$, which is closed with all points being limit points, so it is perfect. On the other hand, $B = [0, 1] \cup \{3\}$ is not perfect because it contains a point 3, which is not a limit point (it is an isolated point).

(i) E is *bounded* if there is a real number M and a point $q \in X$ such that $d(p, q) < M$ for $\forall p \in E$.

(j) E is *dense in* X if every point of X is a limit point of E, or a point of E (or both).

Let us note that in $\mathbb{R}^1$ neighborhoods are segments, whereas in $\mathbb{R}^2$ neighborhoods are interiors of circles.

Theorem 1.5. *Every neighborhood is an open set.*

Proof. Consider neighborhood $E = N_r(p)$, and let q be any point of E. Then there is a positive real number h such that

$$d(p, q) = r - h.$$

For all points s such that $d(q, s) < h$, we have then

$$d(p, s) \leq d(p, q) + d(q, s) < r - h + h = r,$$

so that $s \in E$. Thus, q is an interior point of E. □

Theorem 1.6. *If p is a limit point of a set E, then every neighborhood of p contains infinitely many points of E.*

Proof. Suppose there is a neighborhood N of p which contains only a finite number of points of E. Let $q_1, \ldots, q_n$ be those points of $N \cap E$, which are distinct from p, and put

$$r = \min_{1 \leq m \leq n} d(p, q_m)$$

The minimum of a finite set of positive numbers is clearly positive, so that $r > 0$.

The neighborhood $N_r(p)$ contains no point q of E such that $q \neq p$, so that p is not a limit point of E. This contradiction established the theorem. □

Corollary 1.7. *A finite point set has no limit points.*

Theorem 1.8. *A set E is open if and only if its complement is closed.*

1.2 Compact sets

Definition 1.9. By an *open cover* of a set E in a metric space X we mean a collection $\{G_\alpha\}$ of open subsets of X such that $E \subset \cup_\alpha G_\alpha$.

Definition 1.10. A subset K of a metric space X is said to be *compact* if every open cover of K contains a *finite subcover*. More explicitly, the requirement is that if $\{G_\alpha\}$ is an open cover of K, then there are finitely many indices $\alpha_1, \ldots, \alpha_n$ such that

$$K \subset G_{\alpha_1} \cup \ldots \cup G_{\alpha_n}.$$

Corollary 1.11. *A set E is compact if it is both closed and bounded.*

1.3 Functions

Definition 1.12. Consider two sets A and B, whose elements may be any objects whatsoever, and suppose that with each element x of A there is associated, in some manner, an element of B, which we denote by $f(x)$. Then f is said to be a *function* from A to B (or a *mapping* from A into B). The set A is called the *domain* of f (we also say f is defined on A), and the elements $f(x)$ are called the *values* of f. The set of *all* values of f is called the *range* of f.

Definition 1.13. If for every $y \in B$ there is at most one $x \in A : f(x) = y$, the function f is said to be a 1-1 (*one-to-one*) mapping of A into B. This may also be expressed as follows: f is a 1-1 mapping of A into B provided that $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$, $x_1 \in A$, $x_2 \in A$.

Definition 1.14. Let A and B be two sets and let f be a mapping of A into B. If $f(A) = B$, we say that f maps A *onto* B. If, additionally, f is 1-1, then f is *one-to-one and onto (bijection).*

Definition 1.15. If there exists a 1-1 mapping of A *onto* B, we say that A and B can be put in 1-1 *correspondence*, or that A and B have the same *cardinal number*, or, briefly, that A and B are *equivalent*, and we write $A \sim B$.

Definition 1.16. For any positive integer n, let J_n be the set whose elements are the integers $1, 2, \ldots, n$; let J be the set consisting of all positive integers. For any set A, we say:

(a) A is finite if $A \sim J_n$ for some n.

(b) A is *infinite* if A is not finite.

(c) A is *countable* if $A \sim J$.

(d) A is *uncountable* if A is neither finite nor countable.

(e) A is *at most countable* if A is finite or countable.

For two finite sets A and B, we evidently have $A \sim B$ if and only if A and B contain the same number of elements (same *cardinality*). For infinite sets, however, the idea of cardinality becomes quite vague, whereas the notion of 1-1 correspondence retains its clarity.

Example 1.17. Let A be the set of all integers. Then A is countable. Consider, the following arrangement of the sets A and J:

$$\begin{aligned} A: &\quad 0, 1, -1, 2, -2, \ldots \\ J: &\quad 1, 2, 3, 4, 5, \ldots \end{aligned}$$

We can, in this example, even give an explicit formula for a function f from J to A which sets up a 1-1 correspondence:

$$f(n) = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even,} \\ -\frac{n-1}{2} & \text{if } n \text{ is odd.} \end{cases}$$

Remark 1.18. A finite set cannot be equivalent to one of its proper subsets. That this is, however, possible for infinite sets, is shown by Example 1.17, in which J is a proper subset of A.

Definition 1.19. In the following, assume that the set A is a subset of $\mathbb{R}$.

(a) If there exists $x \in \mathbb{R}$ such that for every $y \in A$ we have $x \geq y$, then the set A is *bounded from above.*

(b) If there exists $x \in \mathbb{R}$ such that for every $y \in A$ we have $x \leq y$, then the set A is *bounded from below.*

(c) The *supremum* of A, denoted as $\sup A$, is the smallest upper bound of the set A.

(d) The *infimum* of A, denoted as $\inf A$, is the largest lower bound of the set A.

We note that the set A is bounded, if it is bounded both from below and from above, which is equivalent to the Definition 1.4(i). If the set A is not bounded from above, then $\sup A = \infty$, and if it is not bounded from below, then $\inf A = -\infty$.

2 Sequences and limits

Definition 2.1. By a *sequence*, we mean a function f defined on the set J of all positive integers. If $f(n) = x_n$ for $n \in J$, it is customary to denote the sequence f by the symbol $\{x_n\}$, or sometimes by $x_1, x_2, x_3, \ldots$. The values of f, that is, the elements x_n, are called the *terms* of the sequence. If A is a set and if $x_n \in A$ for all $n \in J$, then $\{x_n\}$ is said to be a *sequence in* A, or a *sequence of elements of* A.

Note that the terms $x_1, x_2, x_3, \ldots$ of a sequence need not be distinct.

Since every countable set is the range of a 1-1 function defined on J, we may regard every countable set as the range of a sequence of distinct terms. Speaking more loosely, we may say that the elements of any countable set can be "arranged in a sequence".

Definition 2.2. For a given sequence $\{x_n\}$, if $x_{n+1} > x_n$ for $\forall n \in J$, then the sequence is *increasing*. If $x_{n+1} < x_n$ for $\forall n \in J$, then the sequence is *decreasing*. If $x_{n+1} \geq x_n$ for $\forall n \in J$, then the sequence is *non-decreasing*. If $x_{n+1} \leq x_n$ for $\forall n \in J$, then the sequence is *non-increasing*.

If at least one of these four conditions is satisfied, the sequence is called *monotonic*.

Example 2.3. We give examples of different sequences below.

(a) A sequence that is defined via a formula for the nth term: $x_n = \left(\frac{2}{3}\right)^n$.

(b) A sequence that is defined recursively (Fibonacci sequence): $x_n = x_{n-1} + x_{n-2}$ for $n \geq 3$, and $x_1 = x_2 = 1$.

(c) A sequence $x_n = (-1)^n$.

(d) A sequence $x_n = 2^n$.

Note that the sequence (a) is decreasing with n, while the sequence (b) is non-decreasing with n. The sequence (c) is non-monotonic.

Definition 2.4. A sequence $\{x_n\}$ in a metric space X is said to *converge* if there is a point $x \in X$ with the following property: for every $\varepsilon > 0$ there is an integer N such that $n \geq N$ implies that $d(x_n, x) < \varepsilon$.

In this case, we also say that $\{x_n\}$ converges to x, or that x is the limit of $\{x_n\}$, and we write $x_n \to x$, or

$$\lim_{n\to\infty} x_n = x.$$

If $\{x_n\}$ does not converge, it is said to *diverge*.

We recall that the set of all points x_n $(n = 1, 2, 3, \ldots)$ is the *range* of $\{x_n\}$. The range of a sequence may be a finite set, or it may be infinite. The sequence $\{x_n\}$ is said to be *bounded* if its range is bounded. In the Example 2.3, (a) and (c) are bounded sequences, while (b) and (d) are not.

Example 2.5. Show that $\lim_{n\to\infty}(\frac{2}{3})^n = 0$.

We need to show that for a given $\varepsilon > 0$, after some $n \in J$, the distance between the elements of the sequence and the limit 0 is smaller than ε. In other words, that there exists some N such that for all n larger than N we have $d(x_n, 0) < \varepsilon$. Taking the absolute value, we have $|(\frac{2}{3})^n| < \varepsilon$ for $\forall n \geq N$, and rewriting

$$\begin{aligned}
\left(\frac{2}{3}\right)^n &< \varepsilon, \\
\log\left(\frac{2}{3}\right)^n &< \log\varepsilon, \\
n\log\left(\frac{2}{3}\right) &< \log\varepsilon, \\
n &> \frac{\log\varepsilon}{\log 2/3}.
\end{aligned}$$

Denote the smallest integer larger than a as $\lceil a \rceil$. Then, one can take $N = \lceil n \rceil$, and for all $n \geq N$, the inequality $n > \frac{\log\varepsilon}{\log 2/3}$ is satisfied. Then, 0 is a limit of $\left(\frac{2}{3}\right)^n$.

Theorem 2.6. *Every bounded, monotonic sequence converges.*

Example 2.7. Show that the sequence

$$x_n = \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \ldots + \frac{1}{n!} = \sum_{k=1}^{n} \frac{1}{k!}$$

converges.

To show that the sequence converges, we use the Theorem 2.6, hence, it is sufficient to show that the sequence is monotonic and bounded. To show monotonicity, note that

$$x_{n+1} = x_n + \frac{1}{(n+1)!} > x_n,$$

so $\{x_n\}$ is increasing and hence monotonic. To show that it is bounded, note that

$$\frac{1}{n!} = \frac{1}{1 \cdot 2 \cdot 3 \cdot \ldots \cdot n} = \frac{1}{2 \cdot 3 \cdot \ldots \cdot n} \leq \frac{1}{2 \cdot 2 \cdot \ldots \cdot 2} = \frac{1}{2^{n-1}},$$

with strict inequality for $n > 1$. $x_1 = 1$ is finite, hence does not contradict boundedness. For $n > 1$, we have

$$x_n < 1 + \frac{1}{2^1} + \frac{1}{2^2} + \ldots + \frac{1}{2^{n-1}} = \frac{1 - (1/2)^n}{1 - 1/2} = 2 - \left(\frac{1}{2}\right)^{n-1} < 2.$$

Because each element of the sequence x_n for $\forall n > 1$ is bounded by 2, the sequence is bounded.

2.1 Limit laws (i)

Corollary 2.8. *Let $\{x_n\}$ and $\{y_n\}$ are convergent sequences, and let c be a constant. Then,*

(a) $\lim_{n\to\infty} (x_n + y_n) = \lim_{n\to\infty} x_n + \lim_{n\to\infty} y_n$.

(b) $\lim_{n\to\infty} (x_n - y_n) = \lim_{n\to\infty} x_n - \lim_{n\to\infty} y_n$.

(c) $\lim_{n\to\infty} cx_n = c \lim_{n\to\infty} x_n$.

(d) $\lim_{n\to\infty} c = c$.

(e) $\lim_{n\to\infty} (x_n y_n) = \lim_{n\to\infty} x_n \lim_{n\to\infty} y_n$.

(f) $\lim_{n\to\infty} \frac{x_n}{y_n} = \frac{\lim_{n\to\infty} x_n}{\lim_{n\to\infty} y_n}$ *if* $\lim_{n\to\infty} y_n \neq 0$.

(g) $\lim_{n\to\infty} x_n^p = \left(\lim_{n\to\infty} x_n\right)^p$ *if* $p > 0$ *and* $x_n > 0$.

Example 2.9. Find the limit of $\{x_n\}$, where

$$x_n = \frac{2n^3 + n^2 - 7n}{n^3 + 2n + 2}.$$

Rewrite the nth term of the sequence as

$$\frac{2 + n^{-1} - 7n^{-2}}{1 + 2n^{-2} + 2n^{-3}}.$$

The limit of the numerator and the denominator respectively is

$$\lim_{n\to\infty} \left(2 + \frac{1}{n} - \frac{7}{n^2}\right) = 2, \quad \lim_{n\to\infty} \left(1 + \frac{2}{n^2} + \frac{2}{n^3}\right) = 1,$$

so that $\lim_{n\to\infty} x_n = 2$.

Definition 2.10. Given a sequence $\{x_n\}$, consider a sequence $\{n_k\}$ of natural numbers, such that $n_1 < n_2 < n_3 < \ldots$. Then the sequence $\{x_{n_i}\}$ is called a *subsequence* of $\{x_n\}$. If $\{x_{n_i}\}$ converges, its limit is called a *subsequential limit* of $\{x_n\}$.

The sequence $\{x_n\}$ converges to x if and only if every subsequence of $\{x_n\}$ converges to x.

Example 2.11. Consider a sequence $x_n = (-1)^n$ that we know to be divergent. Now, consider two sequences of natural numbers, $\{n_k\} = \{1, 3, 5, \ldots\}$ and $\{m_k\} = \{2, 4, 6, \ldots\}$. The subsequence corresponding to $\{n_k\}$ is $\{-1, -1, -1, \ldots\}$ with the limit -1, and the subsequence corresponding to $\{m_k\}$ is $\{1, 1, 1, \ldots\}$ with the limit 1. Hence, it is possible for subsequences to converge even though the whole sequence does not.

2.2 Upper and lower limits

Definition 2.12. Let $\{x_n\}$ be a sequence of real numbers with the following property: for every real M there is an integer N such that $n \geq N$ implies $x_n \geq M$. We then write

$$x_n \to +\infty.$$

Similarly, if for every real M there is an integer N such that $n \geq N$ implies $x_n \leq M$, we write

$$x_n \to -\infty.$$

Definition 2.13. Let $\{x_n\}$ be a sequence or real numbers. Let E be the set of numbers x such that $x_{n_k} \to x$ for some subsequence $\{x_{n_k}\}$. This set E contains all subsequential limits as defined in the Definition 2.10, plus possibly the numbers $+\infty$, $-\infty$.

Put

$$x^\star = \sup E, \quad x_\star = \inf E.$$

The numbers $x^\star$ and $x_\star$ are called the *upper* and *lower limits* of $\{x_n\}$. We use the notation

$$\limsup_{n\to\infty} x_n = x^\star, \quad \liminf_{n\to\infty} x_n = x_\star.$$

Theorem 2.14. *If $s_n \leq t_n$ for $n \geq N$, where N is fixed, then*

$$\liminf_{n\to\infty} s_n \leq \liminf_{n\to\infty} t_n,$$
$$\limsup_{n\to\infty} s_n \leq \limsup_{n\to\infty} t_n.$$